✅ 1. Competency #7: Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

- Highest relevance.
- This task centers on manipulating an algebraic formula:
  - Students substitute $2n$ for n :

$$D_{\text{new}} = \frac{dv}{60(2n)} = \frac{1}{2} \cdot \frac{dv}{60n} = \frac{1}{2}D$$

- They interpret the structural relationship: doubling the denominator halves the result.
- No numerical computation is needed—only symbolic manipulation and interpretation of functional dependence.
- This is a pure example of working fluently with algebraic expressions and equations, making Competency #7 the most directly engaged.

✅ 2. Competency #12: Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- Highly relevant.
- Although brief, the reasoning involves distinct steps:
  1. Identify the variable being changed ($n \to 2n$);
  2. Substitute into the formula;
  3. Simplify the expression;
  4. Compare new and original values;
  5. Draw a quantitative conclusion ("halved").
- This logical sequence constitutes a concise but complete multi-step mathematical procedure, fulfilling Competency #12.

✅ 3. Competency #8: Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- Explicitly required.
- The prompt asks students to "describe precisely" how D changes.
- A strong response must go beyond "it gets smaller" and state clearly:

> "If the infusion time n is doubled while d and v remain constant, the drip rate D is halved."

- This demands precise verbal communication of a mathematical relationship, directly aligning with Competency #8.

✅ 4. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process

- Moderately relevant.
- The medical context (IV drips) reinforces that:
  - Longer infusion time → slower drip rate (intuitively reasonable);
  - Precision matters (a nurse must calculate correctly for patient safety).
- While the math can be done abstractly, contextual plausibility helps students verify their conclusion (e.g., "Halving makes sense—you don't want to flood the patient over twice as long!").
- Thus, context supports reasonableness checking, though the core reasoning is symbolic.

✅ 5. Competency #1: Perform a simple calculation

- Minimal relevance.
- No arithmetic is required—only symbolic substitution and simplification.
- If a student plugs in numbers (e.g., let $d = 20, v = 600, n = 2$), they perform basic division, but this is optional and not necessary.
- The essence is structural, not computational.

✅ 1. 能力 #7：操作数字、图表与统计信息、代数式、方程以及几何表示

- 最高相关性。
- 本题核心在于操作代数公式：
  - 学生将 n 替换为 $2n$ ：

$$D_{新} = \frac{dv}{60(2n)} = \frac{1}{2} \cdot \frac{dv}{60n} = \frac{1}{2}D$$

- 理解其结构关系：分母加倍，结果减半。
- 无需数值计算——仅需符号操作与函数依赖关系的解释。
- 这是熟练处理代数表达式与方程的典型范例，因此能力 #7 最为核心。

✅ 2. 能力 #12：使用多步骤程序得出数学解、结论或一般化结果

- 高度相关。
- 尽管简短，但推理包含清晰的步骤：

  1. 识别被改变的变量（ $n \to 2n$ ）；
  2. 代入公式；
  3. 化简表达式；
  4. 比较新值与原值；
  5. 得出定量结论（“减半”）。

- 这一逻辑链条构成一个简洁但完整的多步骤数学程序，完全符合能力 #12。

✅ 3. 能力 #8：清晰表述解题过程，展示和/或总结并呈现中间数学结果

- 明确要求。
- 题目要求学生“准确描述” D 的变化。
- 优秀回答不能仅说“变小了”，而应明确指出：

> “若输液时间 n 延长一倍，而 d 和 v 保持不变，则滴速 D 减半。”

- 这要求用精确语言传达数学关系，直接体现能力 #8。

✅ 4. 能力 #13：利用对情境的理解来指导或加速数学解题过程

- 中等相关。
- 医疗情境（静脉输液）强化了以下直觉：
  - 输液时间越长 → 滴速应越慢（符合常识）；
  - 精确性至关重要（护士计算错误可能危及患者安全）。
- 虽然数学可抽象完成，但情境合理性帮助学生验证结论（例如：“减半是合理的——总不能在更长时间内还快速输液！”）。
- 因此，情境支持了结果的合理性检验，尽管核心推理仍是符号化的。

✅ 5. 能力 #1：执行简单计算

- 相关性极低。
- 本题无需算术运算——仅需符号代入与化简。
- 若学生代入具体数值（如设 $d = 20, v = 600, n = 2$），会涉及简单除法，但这非必需。
- 问题本质是结构性的，而非计算性的。